

---

#### 4.8.3 Solar Energy Facilities

- A. Purpose. The purpose of this bylaw is to regulate the development of solar energy facilities by providing standards for the placement, design, construction, operation, monitoring, modification, and removal of such installations, to protect public safety, to minimize impacts on scenic, natural and historic resources of Ware, and to provide adequate financial assurance for the eventual decommissioning of such installations. This purpose is to:
1. facilitate the protection of open space, agricultural land, forested land and mature forest;
  2. facilitate the protection of natural, historic, and scenic resources;
  3. protect endangered or threatened species, wildlife corridors and habitat;
  4. take into consideration sensitive and noteworthy natural, scenic, historic and cultural resources;
  5. identify and protect physical features: streams, wetlands, steep slopes, land forms, cultural assets, and agricultural resources;
- B. Applicability. This §4.8.3 applies to all Large Solar Energy Facilities and to physical modifications that materially alter the type, configuration, or size of these facilities or related equipment. Solar facilities for the primary purpose of agriculture are exempt from this §4.8.3 pursuant to MGL c. 40A §3. Solar facilities for one and two family dwellings are also exempt from this §4.8.3. Solar facilities built as a solar parking canopy are also exempt from §4.8.3, but will require Site Plan Review.
- C. General Requirements
1. Permit Required. Large Solar Energy Facilities are permitted by-right upon approval of a Site Plan in the Highway Commercial (HC), the Commercial Industrial (CI), and the Industrial (I) districts, and upon the granting of a Special Permit and Site Plan Approval in the Residential Business (RB) and the Rural (RQ and RR) districts. Large Solar Energy Facilities are not permitted in the Beaver Lake Residential (BLR), Suburban Residential (SR), Downtown (DTR and DTC), or the Mill Yard (MY) districts, given the density of development in these areas.
  2. Review Fees. The Planning Board may, in accordance with MGL c. 44 §53G, impose a Project Review Fee to hire outside consultants to review the proposed project. This fee shall be estimated at the time of application and a deposit of at least 75% of that estimate collected prior to the public hearing, with the balance due upon receipt of the final invoice from the review consultant. This fee shall be deposited into a separate account in compliance with MGL c. 44 §53G, and any balance (including interest) remaining when the project is completed shall be returned to the applicant.
  3. Site Control
    - a) No more than fourteen (14) total large scale solar arrays shall be permitted within the Town of Ware.

---

b)Utility-Scale Solar Facility

1. No more than one (1) utility-scale solar facility shall be permitted within the Town of Ware. Such facility shall be identified as a large scale solar array over 15MW in total size.
  2. Solar array approvals already permitted as of May 11, 2020 may not be combined with new projects to create a utility-scale solar facility.
  3. Utility-scale solar facility shall be minimally visible from public ways.
  4. If a utility-scale solar facility is proposed to be developed over multiple properties (as recorded at any Massachusetts Registry of Deeds) the dimensional setback and buffer requirements may be reduced at the discretion of the Planning Board. This setback reduction shall only be considered for interior parcel lines where both parcels are being proposed to be developed for a utility-scale solar facility.
  5. A utility-scale solar facility may be excluded from frontage requirements so long as there is legal frontage and access for the project as a whole entity. Frontage and access are not required to be at the same point. Access must be on a public way.
  6. For a utility-scale solar project, a minimum area equal to the total project area must remain as natural open space (i.e. not used in any way by the solar developer) for the life of the project. If the proposed project is to be developed over multiple properties (as recorded at any Massachusetts Registry of Deeds) the fifty percent (50%) open space may be calculated using the total acreage of all parcels included in the project. Nothing in this section shall restrict the landowners from using their property that is subject to a utility-scale solar facility project for other uses, if properly permitted and allowed.
  7. A utility-scale solar facility may be a second primary use on a property where a primary use already exists.
- c) The acreage of a single large-scale solar array site may not exceed 40 acres.
- d) A single large-scale solar array site may not cross property boundaries which have separate ownership.
4. Required Documents, Plans, Drawings & Submissions. In addition to the submission requirements in §7.4, Site Plan Review, the applicant shall provide, with the site plan application, documents, plans, drawings & submissions addressing the below listed criteria:
- a) Signed and stamped plans and drawings, by a Professional Engineer licensed to practice in Massachusetts and in good standing, showing in complete detail the proposed layout of the facility and specifically

---

and completely providing all information required by criteria (b) through (l) below.

- b) Existing conditions showing property lines and physical features, including but not limited to: wetlands and related buffer zones, rivers and associated riverfront areas, vernal pools, flood zones, ledge, outcropping, logging or access roads, forested areas, stone walls and stone foundations or other historic features, and roads for the project site
- c) Locations of local or National Historic Districts
- d) Locations of Priority Habitat Areas and Bio Map 2 Critical Natural Landscape Core Habitat mapped by Natural Heritage & Endangered Special Program (NHESP)
- e) Proposed changes to the site including: grading, slope, vegetation clearing and planting, exterior lighting, boundaries of vegetative buffer, zoning setback lines, screening vegetation or structures
- f) Landscape plan identifying location, size and species of all proposed plantings
- g) Proposed layout of the system and related structures, including final stormwater and other site management devices, fences, emergency management turnarounds and staging areas, and access road(s)
- h) Visual renderings showing sight lines from abutting residential structures, commercial structures and public ways.
- i) Construction stormwater management and erosion control
- j) Post-installation stormwater management
- k) Construction phases with detailed notes on plan including material lay down and staging areas
- l) One- or three- line electrical diagram detailing the solar installation, associated components, and electrical interconnection methods, with all National Electrical Code compliant disconnects and overcurrent devices
- m) Technical specifications of the major system components, including the solar arrays, mounting system, inverter, and any on-site battery storage

---

n) Evidence that the utility company has been notified of the intent to install a solar energy facility and that the utility company has responded in writing to the notice. Off-grid systems are exempt from this requirement

o) Evidence the solar PV modules shall be positioned in such a way that there is minimal or no level of glare on a residence or public way at any time during the day. No solar energy facility shall produce glare that would constitute a nuisance to the public.

p) The name, address, and contact information of the proposed installer, operator, and owner of the solar energy facility

q) Documentation of actual or proposed control of access ways and the project site sufficient to allow for installation and use of the proposed facility

r) An Operation and Maintenance Plan which shall include measures for maintaining safe access, storm water controls, and general procedures for operating and maintaining the installation

s) Proof of liability insurance

t) Financial surety that satisfies §4.8.3 H 5 of this Bylaw

u) A comprehensive decommissioning/removal plan

Any portion of these documentary requirements may be waived for good cause shown upon written request submitted to the Planning Board. Waiver will be granted if the Planning Board determines that there is good cause and that the materials submitted are sufficient for the board to make a decision.

5. Modifications. All major modifications to the site plan that are proposed after site plan approval has been granted require approval by the Planning Board.

6. Dimensional Requirements

a) Setbacks. Large Solar Energy Facilities and all accessory structures shall have a setback from front, side and rear property lines and public ways of at least one-hundred (100') feet.

b) Along property lines abutting land owned by the Department of Fish & Wildlife, Department of Conservation & Recreation, a Land Trust or similar entity, or that is under a Conservation Restriction, solar collectors shall have a setback of two-hundred (200'). This may also include land classified as Chapter 61, 61A, or 61B at the time of the application, particularly if such land is in chapter for the conservation of open space and protection of wildlife habitat. This dimensional setback may be reduced at the discretion of the Planning Board,

---

particularly in locations where the applicant can demonstrate that enforcing this setback would be a hardship to a proposed project and the Board can be shown there is sufficient protection of existing abutting wildlife habitat, such as in the Rural Quabbin district. This may not be reduced to less than the 100' solar setback.

c) Any Large Solar Energy Facilities property along a designated Scenic Road shall have a setback of two-hundred 200'. The Planning Board may require additional buffering along such roads that is consistent with the rural New England character of said roads.

d) Buffer. The site plan shall provide a buffer of one-hundred (100') feet between the solar array and all properties. This buffer shall be of undisturbed natural vegetation, or if existing vegetation is inadequate, as determined by the Planning Board, to provide a visual screen then new landscaping with or without an earthen berm shall be designed and installed to provide the screening. If the applicant provides information showing that the visual buffer would have a detrimental impact on the ability to generate power, the Planning Board may grant a waiver to reduce the size of the buffer, but shall not eliminate it. Consideration should be given to designing buffers in situations where topography may negate the effectiveness of the buffer; the intent is to provide visual screening from existing residences to the array, and as such the location of the buffer can be modified to provide a more effective screen.

D. 7. Accessory Structures. All accessory structures including but not limited to equipment shelters, storage facilities, transformers, battery storage and substations shall be subject to reasonable conditions by the Planning Board concerning bulk and height, parking, building coverage, and vegetative screening to avoid adverse impacts on the neighborhood or abutting properties. Design Standards

1. Lighting

- a) Lighting shall be limited to that required for safety and operational Purposes, and shall not be intrusive in any way on abutting properties.
- b) Lighting shall incorporate full cut-off fixtures to reduce light pollution.

2. Signage

- a) The facility shall have one sign not to exceed 32 square feet which identifies the operator and provides a 24-hour emergency contact phone number.
- b) The site may have a secondary sign not to exceed 32 square feet providing educational information about the facility and the benefits of renewable energy.

3. Utility Connections

- 
- a) The applicant shall place all wiring from the solar installation underground, except in cases where the soil conditions of the site make underground wiring infeasible.
  - b) All utility poles and connections shall be located in the least invasive location as feasibly possible
4. Environmental Impacts
- a) Proposed structures (including solar collectors) shall be integrated into the existing terrain and surrounding landscape by minimizing impact to wetlands, steep slopes, and hilltops; protecting visual amenities and scenic views; minimizing tree, vegetation and soil removal; and minimizing grade change.
  - b) To the greatest extent feasible, solar PV arrays shall be designed to minimize the number of trees removed, the length of stone walls removed, and the area of wetland vegetation disturbed.
  - c) No earth material shall be removed from the site. All cuts and fills of earth material shall be balanced.
  - d) Clearing of natural vegetation shall be limited to what is necessary for the construction, operation and maintenance of the Large Solar Energy Facility
  - e) The topping or pollarding of trees is not allowed in order to promote and protect natural, scenic resources
  - f) For all projects, a minimum area equal to the total project area must remain as natural open space for the life of the project. This land must be on the same lot and contiguous in area. No more than fifty percent (50%) of the lot may be developed for a large scale solar array.
5. Historical Impacts
6. All Large Solar Energy Facility projects shall be reviewed by the Historical Commission. The removal of stonewalls, historic foundations or other historic land features shall be avoided to the greatest extent possible.
7. Prime Farmland Impacts
- a) Prime Farmland may only be developed if co-developed as an area for both solar power as well as for agriculture. Such dual-use solar arrays developed on Prime Farmland shall adhere to construction performance standards as outlined in §4.8.3 as applicable to dual-use operation.
8. Stormwater
- a) Stormwater generated from the project area shall not be allowed to enter any public ways so as to cause an unsafe condition and no stormwater shall be drained of site in excess of the pre-development conditions.
  - b) Stormwater must recharge on site
-

- 
- c) The stormwater management and treatment facility shall be sized to accommodate a 100-year storm event.
  - d) Design shall create an environmentally sensitive site design by incorporating Low Impact Development (LID) techniques, as appropriate.

9. Access

- a) Access roads shall not have a slope greater than 10%. If an access road is to be paved, the slope may be greater if approved by the Fire Chief.
- b) Main access roads shall be at least twelve (12) feet wide to provide adequate access to all main controls, battery storage, or other appurtenant structures
- c) Adequate access must be provided between the solar collectors and perimeter fence in a width of at least eight (8) feet. This shall be a material other than grass, such drive-on pavers with gravel or grass fill, or gravel.
- d) All stream crossings, bridges, and access road shall have the load capacity of fifty (50) tons
- e) Plans must be reviewed by the Ware Fire Department prior to commencement of project.

E. Maintenance

- 1. The operator shall maintain the facility in good condition.
- 2. Maintenance shall include but not be limited to painting, structural repairs, and integrity of security measures.
- 3. The operator shall be responsible for maintaining adequate access for emergency vehicles and maintenance equipment.
- 4. The operator shall be responsible for maintaining safe access, stormwater controls, vegetation controls – including the prompt replacement of any dead/diseased screening trees in the buffer, and general procedures and schedules for operating and maintaining the energy facility.
- 5. Replacement of dead/diseased screening trees must be replaced with a tree of same or similar size in height, width, and condition as the other species in the screening buffer to maintain a continuous buffer.
- 6. If applicable, a battery end-of-life plan shall be included in the Operation & Maintenance Plan – detailing procedures for the removal, disposal, and replacement of retired batteries.
- 7. The owner of the large scale solar array shall file an operation and maintenance report annually with the Planning Board no later than forty-five (45) days after the end of the calendar year, confirming that the operation is ongoing, has not been abandoned, and that the operation and maintenance plan is being followed. Failure to file the report within the required time frame may be considered evidence that the facility has been abandoned.

F. Emergency Services

- 
1. The operator shall provide a copy of the operation and maintenance plan, electrical schematic, and site plan to the Fire Chief.
  2. The operator shall cooperate with local emergency services in developing an emergency response plan; this plan shall be reviewed annually with local emergency officials and revised as necessary. If applicable, this shall include battery storage hazard mitigation.
  3. All means of shutting down the installation shall be clearly marked. Signage on the premises (see §6.5) shall identify a qualified contact person to provide assistance during an emergency; the operator shall change the contact information immediately whenever a change in personnel occurs.
  4. If access to the site is gated and locked, the owner shall provide the Fire Department with a means of entering the property (key, combination, or some other means acceptable to the Fire Department).
  5. Fire Chief will review and approve all site plans for large scale solar arrays to ensure slope of access roads, width of access roads, emergency staging area, and any other emergency protocols needed to accommodate emergency vehicles are addresses.

G. Decommissioning

1. Removal Requirements.
    - a) Any solar installation that is discontinued or abandoned shall be removed in compliance with the requirements of the Building Inspector.
    - b) The owner or operator shall physically remove the installation within 180 days after the date of discontinued operations or receipt of a notice of abandonment from the Town.
  2. Notice to Town.
    - i. Town of Ware officials (Town Manager and Planning Board via the PCD Department) shall be notified by certified mail thirty (30) days prior to any change of ownership of the property, lease holder and/or plant operating company within thirty (30) days of such notice, or at a mutually-convenient time, the new party shall arrange to meet with the Planning Board to review the operating conditions of the Special Permit, or as soon as convenient to both parties
    - ii. In the event a Large Solar Energy Facility is anticipated to be out of service for a period of at least one year, written notice shall be sent by certified mail to the Planning Board, Building Inspector, and Fire Chief stating the anticipated period of non-operation as well as the reason for the shut-down.
    - iii. In the event of decommissioning, the notice shall provide the anticipated schedule for removal and site restoration along with a site plan indicating the site conditions after the decommissioning is completed, including topography at the same contour interval that was provided in the initial site plan approval, if topographical changes will be made.
  3. Abandonment.
    - i. Absent notice of a proposed date of decommissioning or written notice of extenuating circumstances, the solar facility shall be
-



---

considered abandoned when it fails to operate for more than one year. Upon determining that the facility has not been in operation for more than one year, the Building Inspector shall send a written notice by certified mail to the property owner and the last known solar energy facility operator, with a copy to the Planning Board that the Town has found the facility to be abandoned. If the owner or operator fails to remove the installation in accordance with the requirements of this section, the Town may, to the extent it is otherwise duly authorized by law, enter the property and physically remove the facility.

4. Decommissioning shall consist of:
  - i. Physical removal from the site of the solar arrays, structures, equipment, security barriers, and electrical transmission lines.
  - ii. Stabilization or re-vegetation of the site as necessary to minimize erosion. The Building Inspector may allow the owner or operator to leave landscaping, below-grade foundations, and/or access roads in place in order to minimize erosion and disruption of vegetation.
5. Financial Surety.
  - i. Prior to commencing operation, the applicant shall provide a form of surety, either through a cash deposit, in an amount determined to be adequate by the Planning Board to cover the cost of removal and site restoration.
  - ii. The applicant shall submit a fully inclusive estimate of the costs associated with removal, prepared by a qualified engineer. The amount shall include an escalator for calculating increased removal costs due to inflation. Salvage for solar panels shall not be included in decommissioning calculations. Salvage may be included for other parts of the installation at the discretion of the Planning Board.
  - iii. The surety shall be maintained by the developer for the lifespan of the facility, with annual certification notices from the surety company or bank for surety bonds submitted to the Planning Board. Such surety is not required for municipal facilities.